Supplementary Material for "The Optimizer Quotient and the Certification Trilemma"

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1 Full Lean Handle Ledger

This supplement provides the complete Lean handle ledger cited by the manuscript. It includes every handle identifier, declaration name, and source module path.

ID	Lean Handle / Source	ID	Lean Handle / Source
AB2	DecisionProblem.surjective_abstraction_ factors_or_erases DecisionQuotient/AbstractionCollapse.lean	CC4	DecisionQuotient.ClaimClosure.anchor_sigma2p_ complete_conditional DecisionQuotient/ClaimClosure.lean
CR1	DecisionQuotient.ConfigReduction.config_ sufficiency_iff_behavior_preserving DecisionQuotient/Hardness/ConfigReduction.lean	CT12	DecisionQuotient.Physics.ClaimTransport. physical_bridge_bundle DecisionQuotient/Physics/ClaimTransport.lean
DC19	StochasticSequential.stochastic_anchor_ sufficient_of_stochastic_sufficient DecisionQuotient/StochasticSequential/ Quotient.lean	DC23	StochasticSequential.sequential_anchor_ sufficient_of_sequential_sufficient DecisionQuotient/StochasticSequential/ Basic.lean
DC28	StochasticSequential.reduceTQBF_correct_anchor DecisionQuotient/StochasticSequential/ PolynomialReduction.lean	DC29	StochasticSequential.reduceTQBF_to_sequential_ anchor_reduction DecisionQuotient/StochasticSequential/ PolynomialReduction.lean
DC40	StochasticSequential.reduceMAJSATPureAnchor_ correct DecisionQuotient/StochasticSequential/ PolynomialReduction.lean	DC41	StochasticSequential.reduceMAJSAT_to_pure_ stochastic_anchor_reduction DecisionQuotient/StochasticSequential/ PolynomialReduction.lean
DC42	StochasticSequential.stochastic_anchor_check_ pp_hard DecisionQuotient/StochasticSequential/ PolynomialReduction.lean	DC44	StochasticSequential.sequential_anchor_check_ pspace_hard DecisionQuotient/StochasticSequential/ PolynomialReduction.lean
DC45	StochasticSequential.stochastic_sufficiency_ pp_hard DecisionQuotient/StochasticSequential/ PolynomialReduction.lean	DC46	StochasticSequential.sequential_sufficiency_ pspace_hard DecisionQuotient/StochasticSequential/ PolynomialReduction.lean
DC47	StochasticSequential.stochastic_minimum_ sufficiency_pp_hard DecisionQuotient/StochasticSequential/ PolynomialReduction.lean	DC48	StochasticSequential.sequential_minimum_ sufficiency_pspace_hard DecisionQuotient/StochasticSequential/ PolynomialReduction.lean
DC49	StochasticSequential. sequentialMinimalSufficient_iff_relevant DecisionQuotient/StochasticSequential/ Basic.lean	DC50	StochasticSequential.sequentialRelevantSet_is_ minimal DecisionQuotient/StochasticSequential/ Basic.lean
DC57	StochasticSequential.fiberDecisionProblem_ sufficient DecisionQuotient/StochasticSequential/ Basic.lean	DC62	StochasticSequential. countedStochasticSufficiencySearch_spec DecisionQuotient/StochasticSequential/ Computation.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
DC63	StochasticSequential. countedStochasticSufficiencySearch_steps DecisionQuotient/StochasticSequential/ Computation.lean	DC64	StochasticSequential. countedStochasticAnchorSearch_spec DecisionQuotient/StochasticSequential/ Computation.lean
DC65	StochasticSequential. countedStochasticAnchorSearch_steps DecisionQuotient/StochasticSequential/ Computation.lean	DC66	StochasticSequential. countedSequentialSufficiencySearch_spec DecisionQuotient/StochasticSequential/ Computation.lean
DC67	StochasticSequential. countedSequentialSufficiencySearch_steps DecisionQuotient/StochasticSequential/ Computation.lean	DC68	StochasticSequential. countedSequentialAnchorSearch_spec DecisionQuotient/StochasticSequential/ Computation.lean
DC69	StochasticSequential. countedSequentialAnchorSearch_steps DecisionQuotient/StochasticSequential/ Computation.lean	DC70	DecisionQuotient.static_sufficiency_inP_ explicit DecisionQuotient/ExplicitStateMembership.lean
DC71	DecisionQuotient.static_anchor_inP_explicit DecisionQuotient/ExplicitStateMembership.lean	DC72	StochasticSequential.stochastic_sufficiency_ inP_explicit DecisionQuotient/StochasticSequential/ Computation.lean
DC73	StochasticSequential.stochastic_anchor_inP_ explicit DecisionQuotient/StochasticSequential/ Computation.lean	DC74	StochasticSequential.sequential_sufficiency_ inP_explicit DecisionQuotient/StochasticSequential/ Computation.lean
DC75	StochasticSequential.sequential_anchor_inP_ explicit DecisionQuotient/StochasticSequential/ Computation.lean	DC76	DecisionQuotient.explicit_state_inP_summary DecisionQuotient/ExplicitStateMembership.lean
DC80	DecisionQuotient.staticSufficiency_counted_ search_witness DecisionQuotient/AlgorithmComplexity.lean	DC81	DecisionQuotient.static_query_search_matrix DecisionQuotient/AlgorithmComplexity.lean
DC82	DecisionQuotient.finite_search_summary DecisionQuotient/ComplexityMain.lean	DC91	StochasticSequential. countedStochasticPreservationSearch_spec DecisionQuotient/StochasticSequential/ Computation.lean
DC92	StochasticSequential. countedStochasticPreservationSearch_steps DecisionQuotient/StochasticSequential/ Computation.lean	DC93	StochasticSequential.stochastic_preservation_ inP_explicit DecisionQuotient/StochasticSequential/ Computation.lean
DC94	StochasticSequential.stochastic_preservation_ implies_static_sufficiency DecisionQuotient/StochasticSequential/ Basic.lean	DC95	StochasticSequential.static_sufficiency_ implies_stochastic_preservation_of_full_ support DecisionQuotient/StochasticSequential/ Basic.lean
DC96	StochasticSequential.static_sufficiency_iff_ stochastic_preservation_of_full_support DecisionQuotient/StochasticSequential/ Basic.lean	DC97	StochasticSequential.stochasticDecisionEquiv_ iff_decisionEquiv_of_preservation DecisionQuotient/StochasticSequential/ Quotient.lean
DC98	StochasticSequential.stochasticDecisionEquiv_ iff_decisionEquiv_of_full_support DecisionQuotient/StochasticSequential/ Quotient.lean	DC99	StochasticSequential.stochasticEquivSetoid_eq_ decisionSetoid_of_full_support DecisionQuotient/StochasticSequential/ Quotient.lean
DC100	StochasticSequential.benchmark_escalation_ summary DecisionQuotient/StochasticSequential/ Hierarchy.lean	DN1	DecisionProblem.stateDecisionNoise_iff_same_ quotient DecisionQuotient/DecisionNoise.lean
DN2	DecisionProblem.decisionNoise_iff_not_relevant DecisionQuotient/DecisionNoise.lean	DN5	DecisionProblem.decisionNoise_iff_condIndep DecisionQuotient/DecisionNoise.lean
DP1	DecisionQuotient.DecisionProblem. minimalSufficient_iff_relevant DecisionQuotient/Sufficiency.lean	DP2	DecisionQuotient.DecisionProblem.relevantSet_ is_minimal DecisionQuotient/Sufficiency.lean
DP6	ClaimClosure.DP6 DecisionQuotient/ClaimClosure.lean	DP7	ClaimClosure.DP7 DecisionQuotient/ClaimClosure.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
DQ1	ClaimClosure.DQ1 DecisionQuotient/ClaimClosure.lean	DQ2	ClaimClosure.DQ2 DecisionQuotient/ClaimClosure.lean
DQ3	ClaimClosure.DQ3 DecisionQuotient/ClaimClosure.lean	DQ6	ClaimClosure.DQ6 DecisionQuotient/ClaimClosure.lean
DQ7	ClaimClosure.DQ7 DecisionQuotient/ClaimClosure.lean	DQ9	DecisionQuotient.HardnessDistribution. simplicityTax_conservation DecisionQuotient/HardnessDistribution.lean
DQ10	DecisionQuotient.IntegrityCompetence. alwaysAbstain_integrity DecisionQuotient/IntegrityCompetence.lean	DQ11	DecisionQuotient.SetCoverInstance.min_ sufficient_iff_set_cover DecisionQuotient/Hardness/ SetCoverReduction.lean
DQ12	DecisionQuotient.StochasticSequential.bounded_ horizon_tractable DecisionQuotient/StochasticSequential/ Tractability.lean	DQ13	DecisionQuotient.StochasticSequential.bounded_ support_tractable DecisionQuotient/StochasticSequential/ Tractability.lean
DQ14	DecisionQuotient.StochasticSequential.fully_ observable_tractable DecisionQuotient/StochasticSequential/ Tractability.lean	DQ15	DecisionQuotient.StochasticSequential.product_ distribution_tractable DecisionQuotient/StochasticSequential/ Tractability.lean
DQ16	DecisionQuotient.StochasticSequential.product_ enables_transfer DecisionQuotient/StochasticSequential/ CrossRegime.lean	DQ18	DecisionQuotient.StochasticSequential.static_ simpler_than_stochastic DecisionQuotient/StochasticSequential/ CrossRegime.lean
DQ19	DecisionQuotient.StochasticSequential. stochastic_simpler_than_sequential DecisionQuotient/StochasticSequential/ CrossRegime.lean	DQ20	DecisionQuotient.all_coordinates_necessary_of_ not_tautology DecisionQuotient/Hardness/ MinSufficientApproximation.lean
DQ25	DecisionQuotient.counted_min_sufficient_ inapproximability_conditional DecisionQuotient/Hardness/ ApproximationHardness.lean	DQ26	DecisionQuotient.dichotomy_conditional DecisionQuotient/ClaimClosure.lean
DQ27	DecisionQuotient.eth_lower_bound_informal DecisionQuotient/Hardness/ETH.lean	DQ28	DecisionQuotient.exact_certainty_inflation_ under_hardness_core DecisionQuotient/ClaimClosure.lean
DQ30	DecisionQuotient.hard_family_all_coords_core DecisionQuotient/ClaimClosure.lean	DQ31	DecisionQuotient.integrity_resource_bound_for_ sufficiency DecisionQuotient/ClaimClosure.lean
DQ32	DecisionQuotient.min_sufficient_factor_ inapprox_of_set_cover_factor_inapprox DecisionQuotient/Hardness/ ApproximationHardness.lean	DQ34	DecisionQuotient.min_sufficient_set_cover_ equiv DecisionQuotient/Hardness/ ApproximationHardness.lean
DQ36	DecisionQuotient.minsuff_conp_complete_ conditional DecisionQuotient/ClaimClosure.lean	DQ38	DecisionQuotient.no_exact_claim_admissible_ under_hardness_core DecisionQuotient/ClaimClosure.lean
DQ39	DecisionQuotient.singleton_gate_sufficient_of_ tautology DecisionQuotient/Hardness/ MinSufficientApproximation.lean	DQ40	DecisionQuotient.sufficiency_conp_complete_ conditional DecisionQuotient/ClaimClosure.lean
DQ42	DecisionQuotient.sufficient_means_factorizable DecisionQuotient/Information.lean	DQ43	DecisionQuotient.tautology_decidable_from_ factor_approx DecisionQuotient/Hardness/ MinSufficientApproximation.lean
DQ44	DecisionQuotient.tractable_subcases_ conditional DecisionQuotient/ClaimClosure.lean	DQ45	DecisionQuotient.tractable_tree_core DecisionQuotient/ClaimClosure.lean
DQ51	DecisionQuotient.bounded_actions_tractable DecisionQuotient/Summary.lean	DQ56	DecisionQuotient.orbitType_count_bound DecisionQuotient/Tractability/Dimensional.lean
DQ57	DecisionQuotient.separable_utility_tractable DecisionQuotient/Summary.lean	DQ58	DecisionQuotient.stochastic_objective_bridge_ can_fail_on_sufficiency DecisionQuotient/ClaimClosure.lean
DQ62	DecisionQuotient.transition_coupled_bridge_ can_fail_on_sufficiency DecisionQuotient/ClaimClosure.lean	DQ63	DecisionQuotient.tree_structure_tractable DecisionQuotient/Summary.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
DQ72	DecisionQuotient.bounded_actions_complexity DecisionQuotient/Tractability/ BoundedActions.lean	DQ74	DecisionQuotient.sufficiency_poly_separable DecisionQuotient/Tractability/ SeparableUtility.lean
DQ75	DecisionQuotient.tensor_contraction_tractable DecisionQuotient/Tractability/ SeparableUtility.lean	DQ76	DecisionQuotient.low_rank_utility_admits_ factored_computation DecisionQuotient/Tractability/ SeparableUtility.lean
DQ77	DecisionQuotient.low_rank_tractability DecisionQuotient/Tractability/ SeparableUtility.lean	DQ78	DecisionQuotient.sufficiency_poly_tree_ structured DecisionQuotient/Tractability/ TreeStructure.lean
DQ79	DecisionQuotient.csp_treewidth_tractable DecisionQuotient/Tractability/ TreeStructure.lean	DQ80	DecisionQuotient.sufficiency_reduces_to_ interaction_csp DecisionQuotient/Tractability/ TreeStructure.lean
DQ81	DecisionQuotient.bounded_treewidth_ tractability DecisionQuotient/Tractability/ TreeStructure.lean	DQ82	DecisionQuotient.orbitType_eq_iff DecisionQuotient/Tractability/Dimensional.lean
DQ83	DecisionQuotient.symmetric_optimalActions_ orbit_invariant DecisionQuotient/Tractability/Dimensional.lean	DQ84	DecisionQuotient.sufficiency_reduces_to_cross_ orbit_check DecisionQuotient/Tractability/Dimensional.lean
DQ85	DecisionQuotient.symmetric_sufficiency_ complexity_bound DecisionQuotient/Tractability/Dimensional.lean	DQ86	DecisionQuotient.StochasticSequential. countedStochasticAnchorPreservationSearch_spec DecisionQuotient/StochasticSequential/ PreservationVariants.lean
DQ87	DecisionQuotient.StochasticSequential. countedStochasticAnchorPreservationSearch_ steps DecisionQuotient/StochasticSequential/ PreservationVariants.lean	DQ88	DecisionQuotient.StochasticSequential. countedStochasticMinimumPreservationSearch_ spec DecisionQuotient/StochasticSequential/ PreservationVariants.lean
DQ89	DecisionQuotient.StochasticSequential. countedStochasticMinimumPreservationSearch_ steps DecisionQuotient/StochasticSequential/ PreservationVariants.lean	DQ90	DecisionQuotient.StochasticSequential.static_ anchor_implies_stochastic_anchor_preservation_ of_positive_anchor_fiber DecisionQuotient/StochasticSequential/ PreservationVariants.lean
DQ91	DecisionQuotient.StochasticSequential.static_ sufficiency_implies_stochastic_preservation_ of_positive_fiber_support DecisionQuotient/StochasticSequential/ PreservationVariants.lean	DQ92	DecisionQuotient.StochasticSequential. stochasticAnchorPreservation_counted_search_ witness DecisionQuotient/StochasticSequential/ PreservationVariants.lean
DQ93	DecisionQuotient.StochasticSequential. stochasticMinimumPreservation_counted_search_ witness DecisionQuotient/StochasticSequential/ PreservationVariants.lean	DQ94	DecisionQuotient.StochasticSequential. stochastic_anchor_preservation_iff_static_ anchor_of_full_support DecisionQuotient/StochasticSequential/ PreservationVariants.lean
DQ95	DecisionQuotient.StochasticSequential. stochastic_minimum_preservation_iff_static_ of_full_support DecisionQuotient/StochasticSequential/ PreservationVariants.lean	DQ96	DecisionQuotient.StochasticSequential. stochastic_minimum_preservation_static_ relevant_card_le DecisionQuotient/StochasticSequential/ PreservationVariants.lean
DQ97	DecisionQuotient.StochasticSequential. stochastic_preservation_contains_static_ relevant DecisionQuotient/StochasticSequential/ PreservationVariants.lean	DQ98	DecisionQuotient.anchor_sufficiency_sigma2p DecisionQuotient/Hardness.lean
EH1	StochasticSequential.existential_anchor_ source_fits_np_over_ppstyle_honest DecisionQuotient/StochasticSequential/ ExistentialHardness.lean	EH2	StochasticSequential.existential_anchor_hard_ honest DecisionQuotient/StochasticSequential/ ExistentialHardness.lean

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ID	Lean Handle / Source	ID	Lean Handle / Source
EH3	StochasticSequential.existential_anchor_query_family_hard_honest DecisionQuotient/StochasticSequential/ExistentialHardness.lean	EH4	StochasticSequential.existential_anchor_np_over_ppstyle_hard_honest DecisionQuotient/StochasticSequential/ExistentialHardness.lean
EH5	StochasticSequential.existential_anchor_query_family_np_over_ppstyle_hard_honest DecisionQuotient/StochasticSequential/ExistentialHardness.lean	EH6	StochasticSequential.existential_anchor_query_family_np_over_ppstyle_complete_honest DecisionQuotient/StochasticSequential/ExistentialHardness.lean
EH7	StochasticSequential.existential_decisiveness_complement_np_over_ppstyle_hard_honest DecisionQuotient/StochasticSequential/ExistentialHardness.lean	EH8	StochasticSequential.existential_decisiveness_complement_np_over_ppstyle_complete_honest DecisionQuotient/StochasticSequential/ExistentialHardness.lean
EH9	StochasticSequential.existential_decisiveness_query_family_np_over_ppstyle_hard_honest DecisionQuotient/StochasticSequential/ExistentialHardness.lean	EH10	StochasticSequential.existential_decisiveness_query_family_np_over_ppstyle_complete_honest DecisionQuotient/StochasticSequential/ExistentialHardness.lean
EX1	Examples.pomdp_reduction_to_preservation DecisionQuotient/Examples/PreservationExamples.lean	EX2	Examples.hyperparam_reduction_to_static DecisionQuotient/Examples/PreservationExamples.lean
EX3	Examples.toy_full_opt_s1_contains_a DecisionQuotient/Examples/ConcreteExamples.lean	HD14	DecisionQuotient.HardnessDistribution.linear_lt_exponential_plus_constant_eventually DecisionQuotient/HardnessDistribution.lean
HD23	DecisionQuotient.HardnessDistribution.hardness_is_irreducible_required_work DecisionQuotient/HardnessDistribution.lean	IC30	DecisionQuotient.IntegrityCompetence.exactCertaintyInflation_iff_no_exact_competence DecisionQuotient/IntegrityCompetence.lean
IC32	DecisionQuotient.IntegrityCompetence.integrity_forces_abstention DecisionQuotient/IntegrityCompetence.lean	IC33	DecisionQuotient.IntegrityCompetence.integrity_not_competent_of_nonempty_scope DecisionQuotient/IntegrityCompetence.lean
IC34	DecisionQuotient.IntegrityCompetence.integrity_resource_bound DecisionQuotient/IntegrityCompetence.lean	OU1	StochasticSequential.stochastic_anchor_query_fits_np_over_ppstyle DecisionQuotient/StochasticSequential/OracleUpperBounds.lean
OU2	StochasticSequential.stochastic_minimum_query_fits_np_over_ppstyle DecisionQuotient/StochasticSequential/OracleUpperBounds.lean	OU6	StochasticSequential.stochastic_anchor_inP_explicit_via_witness_schema DecisionQuotient/StochasticSequential/OracleUpperBounds.lean
OU7	StochasticSequential.stochastic_minimum_inP_explicit_via_witness_schema DecisionQuotient/StochasticSequential/OracleUpperBounds.lean	OU8	StochasticSequential.fitsCoNPOverPPStyle_of_no_witness DecisionQuotient/StochasticSequential/OracleUpperBounds.lean
OU9	StochasticSequential.stochastic_decisiveness_query_fits_conp_over_ppstyle DecisionQuotient/StochasticSequential/OracleUpperBounds.lean	OU10	StochasticSequential.stochastic_decisiveness_complement_fits_np_over_ppstyle DecisionQuotient/StochasticSequential/OracleUpperBounds.lean
OU11	StochasticSequential.stochastic_decisiveness_scoped_oracle_bounds DecisionQuotient/StochasticSequential/OracleUpperBounds.lean	OU12	StochasticSequential.stochastic_preservation_explicit_summary DecisionQuotient/StochasticSequential/OracleUpperBounds.lean
PA3	Physics.AnchorChecks.stochastic_anchor_check_iff_exists_anchor_singleton DecisionQuotient/Physics/AnchorChecks.lean	QT1	DecisionProblem.quotient_is_coarsest DecisionQuotient/Quotient.lean
QT7	DecisionProblem.quotient_has_unique_factorization DecisionQuotient/Quotient.lean	WD1	DecisionQuotient.checking_witnessing_duality_budget DecisionQuotient/WitnessCheckingDuality.lean
WD2	DecisionQuotient.no_sound_checker_below_witness_budget DecisionQuotient/WitnessCheckingDuality.lean	WD3	DecisionQuotient.checking_time_ge_witness_budget DecisionQuotient/WitnessCheckingDuality.lean

2 Claim-to-Lean Handle Mapping

This section maps each paper claim to its corresponding Lean formalization.

Paper claim	Lean handle
Corollary X.1: No General-Purpose Exact Configuration Minimizer	DQ36
Corollary VII.4: Cross-Model Contrast for Exact Certification	DQ3, DQ26
Corollary IX.9: Exact Reliability Impossibility in the Hard Regime	DQ2, IC32
Proposition X.3: Exact Certification Competence Depends on Regime	DQ36, DQ40, DQ44
Definition II.5: Decision Problem	DP6
Definition VII.3: Exponential Time Hypothesis (ETH)	DQ27, DQ30
Reduction to a static preservation check	EX2
Proposition II.6: Insufficiency Equals Counterexample Witness	DP7
Definition II.9: Minimal Sufficient Set	DP1, DP2
Definition II.14: Exact Relevance Identifiability	DQ6, DQ7
One-step POMDP	EX1
Proposition V.9: Sequential Sufficiency Refines to Sequential Anchor Sufficiency	DC23
Proposition V.12: Explicit Finite Search for Sequential Queries	DC76, DC68, DC69, DC66, DC67, DC75, DC74
Definition V.7: Sequential Decision Problem	DC49, DC50
Proposition V.13: Tractable Sequential Cases	DQ12, DQ14, DQ45
Proposition IV.28: Static Sufficiency Does Not Imply Stochastic Sufficiency	DQ58
Proposition XI.1: Static Sufficiency as Reduct Preservation for the Induced Decision Table	DP1, DP2
Proposition IV.29: Static Singleton Sufficiency Transfers to Stochastic Sufficiency	DQ15, DQ16
Proposition IV.22: Stochastic Decisiveness Yields Stochastic Anchor Sufficiency	DC19
Proposition IV.25: Explicit Finite Search for Stochastic Queries	DC76, DC64, DC65, DC91, DC92, DC62, DC63, DC73, DC93, DC72
Proposition IV.9: Full-Support Equivalence	DC96, DC95
Proposition V.14: Stochastic Sufficiency Does Not Imply Sequential Sufficiency	DQ62
Proposition IV.16: General-Distribution Obstructions and Support-Sensitive Bridge	DQ90, DQ91, DQ96, DQ97
Proposition IV.14: Explicit Finite Search for Preservation Variants	DQ86, DQ87, DQ88, DQ89
Proposition IV.10: Quotient Equivalence under Full Support	DC98, DC97, DC99
Proposition IV.8: Stochastic Sufficiency Implies Static Sufficiency	DC94
Proposition IV.27: Tractable Stochastic Cases	DQ13, DQ15
Proposition II.18: Sufficiency Characterization	DQ42
Theorem III.6: ANCHOR-SUFFICIENCY is Sigma-2-P-complete	CC4, DQ98
Definition VIII.3: Bounded Action Problem	DQ72, DQ51
Theorem VII.1: Encoding-Sensitive Contrast	DQ26, DQ30
Definition IX.6: Certifying Solver	IC34, DQ31

Paper claim	Lean handle
Theorem III.5: Static Collapse Theorem	`DQ36`
Definition VIII.17: Dimensional State Space	`DQ82`
Theorem II.15: Universal Property of the Optimizer Quotient	`QT7`, `QT1`
Theorem VI.1: Regime-Sensitive Complexity Matrix	`DQ1`, `DQ18`, `DQ19`, `DC100`, `EH4`, `EH6`, `EH5`, `EH8`, `EH7`, `EH10`, `EH9`, `OU1`, `OU11`, `OU2`, `OU12`
Definition VIII.5: Separable Utility	`DQ57`, `DQ74`
Theorem V.10: Sequential Anchor Query is PSPACE-complete	`DC28`, `DC29`, `DC44`
Theorem V.11: Sequential Minimum Query is PSPACE-complete	`DC48`, `DC46`
Theorem V.8: Sequential Sufficiency is PSPACE-complete	`DC66`, `DC74`, `DC46`
Theorem VI.2: Static-to-Stochastic Complexity Gap	`DQ18`
Theorem IV.23: Stochastic Anchor Sufficiency is PP-hard	`DC40`, `DC41`, `DC42`
Theorem IV.24: Stochastic Minimum-Sufficient-Set is PP-hard	`DC47`, `DC45`
Theorem IV.26: Stochastic Anchor and Minimum Queries Lie in $\mathsf{NP}^{\mathsf{PP}}$	`PA3`, `EH2`, `EH4`, `EH3`, `EH5`, `EH1`, `OU6`, `OU1`, `OU7`, `OU2`
Definition IV.21: Stochastic Anchor Sufficiency	`DC62`, `OU8`, `OU10`, `OU9`, `OU11`, `DC72`, `DC45`
Definition IV.7: Stochastic Decision Problem	`DC91`, `DC92`, `OU12`, `DC93`
Theorem IV.13: Full-Support Inheritance for Preservation Variants	`DQ94`, `DQ95`
Theorem IV.15: Explicit-State Tractability for Preservation Variants	`DQ92`, `DQ93`
Theorem VI.3: Stochastic-to-Sequential Complexity Gap	`DQ19`
Theorem III.4: SUFFICIENCY-CHECK is coNP-complete	`DQ40`
Theorem VIII.19: Symmetric Complexity Bound	`DQ56`, `DQ85`
Theorem VIII.18: Symmetric Sufficiency Reduction	`DQ84`, `DQ83`
Definition VIII.8: Tensor Rank Decomposition	`DQ77`, `DQ76`, `DQ75`
Theorem VIII.1: Tractable Subcases	`DQ81`, `DQ77`, `DQ85`, `DQ63`
Definition VIII.10: Tree-Structured Dependencies	`DQ78`, `DQ63`
Definition VIII.13: Interaction Graph	`DQ81`, `DQ79`, `DQ80`
Theorem III.7: Witness-Checking Duality	`WD3`, `WD1`, `WD2`
Definition X.5: Slot-Inspection Checker	`WD3`, `WD1`, `WD2`

Auto summary: mapped 58/58 (full=58, derived=0, unmapped=0).